

Input:

Claim, Beliefs, Demographic → Prediction

(1) Utility

Claim + Demo_A → Prediction_A

$\Updownarrow$ demographic swap

Claim + Demo_B → Prediction_B

Metric: Flip rate

Interpretation: Sensitivity may reflect real correlations

(2) Shortcut Reliance

Claims where $P(\text{label} \mid \text{Demo_A}) = P(\text{label} \mid \text{Demo_B})$

Claim + Demo_A → Prediction_A

$\Updownarrow$ demographic swap

Claim + Demo_B → Prediction_B

Metric: Flip rate

Interpretation: Demographic shortcut usage

(3) Complementarity

Randomly drop 70% belief information

Claim + Incomplete Beliefs → Prediction_A

Claim + Incomplete Beliefs + Demo → Prediction_B

Metric: Flip rate, Accuracy

Interpretation: Helpful prior vs harmful steering